

CSV Import Enhancements

Value Mapping and Transformations

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Chapter 1. Executive Summary

This document proposes enhancements to the Quantum framework's CSV import functionality to add value mapping and transformation capabilities. The primary gaps addressed are:

1. **No value mapping** - Cannot map CSV values to different values prior to validation (e.g., "Y" → `true`, "Active" → `ACTIVE`)
2. **No lookup/reference resolution** - Cannot resolve foreign references by looking up values in other collections
3. **No pre-validation transformation hook** - Cannot programmatically transform parsed beans before validation
4. **No explicit intent control** - Cannot specify INSERT vs UPDATE per row; always uses auto-detect (UPSERT)

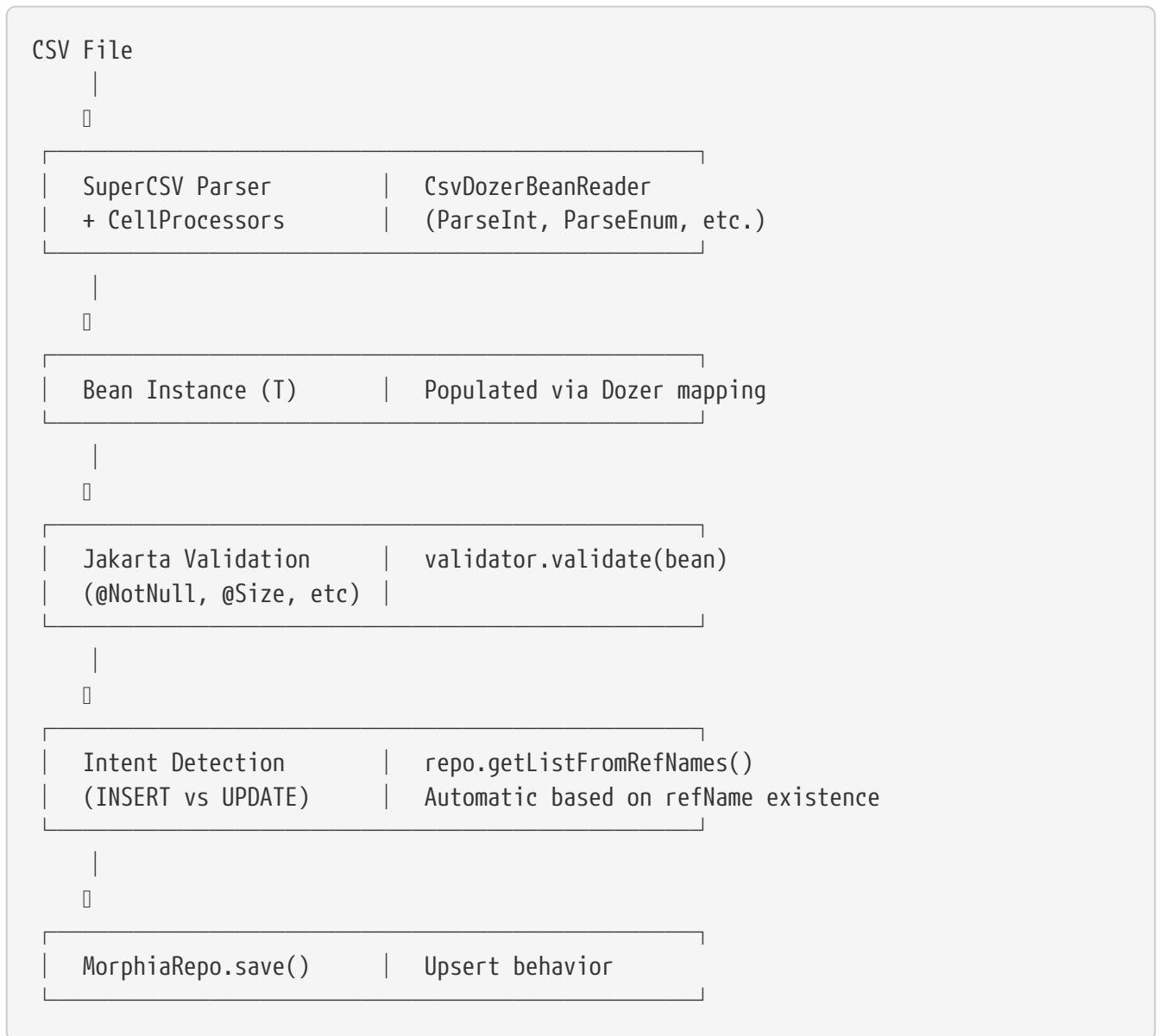


This design intentionally **excludes** MERGE and DELETE intents for safety reasons. Only INSERT, UPDATE, UPSERT, and SKIP are supported.

1. **No header-based field metadata** - Cannot specify required/optional/calculated fields via CSV header conventions

Chapter 2. Current State Analysis

2.1. Existing Architecture



2.2. Current Limitations

Gap	Current Behavior	Desired Behavior
Value Mapping	Only type conversion (String → int, String → enum)	Arbitrary value transformations via configurable mappings
Lookup Resolution	Not available	Resolve references by looking up values in other collections
String Transformations	None	Trim, case conversion, regex replacement
Pre-validation Hook	None	Transform bean after parsing, before validation

Gap	Current Behavior	Desired Behavior
Default Values	None (null if empty)	Configurable default values for empty/null cells
Intent Control	Always UPSERT (auto-detect INSERT vs UPDATE)	Optional per-row intent column (INSERT, UPDATE, SKIP) with validation
Header-based Field Metadata	None - all fields treated equally	Suffix conventions for required (*), optional (?), calculated (~) fields

Chapter 3. Proposed Design

3.1. Overview

The design introduces three new concepts:

1. **ImportProfile** - Configurable transformation and mapping rules
2. **Enhanced CellProcessors** - Value mapping, lookup, and string transformation processors
3. **PreValidationTransformer** - Programmatic transformation hook

CSV File + ImportProfile

|

□

SuperCSV Parser

+ Enhanced Processors

|

← ValueMapProcessor, LookupProcessor

|

□

Bean Instance (T)

|

|

□

PreValidationTransform

|

← NEW: Programmatic transformation

|

□

Jakarta Validation

|

|

□

Intent Detection

|

(unchanged)

|

□

MorphiaRepo.save()

|

(unchanged)

3.2. Component Design

3.2.1. 1. ImportProfile Entity

A persistent, reusable configuration for import transformations.

```
package com.e2eq.framework.model.persistent.imports;

import com.e2eq.framework.model.persistent.base.BaseModel;
import dev.morphia.annotations.Entity;
import io.quarkus.runtime.annotations.RegisterForReflection;
import lombok.Data;
import lombok.EqualsAndHashCode;
import lombok.NoArgsConstructor;
import lombok.experimental.SuperBuilder;

import java.util.List;

/**
 * Persistent configuration for CSV import transformations.
 * Profiles can be reused across multiple imports and shared within a tenant.
 */
@Data
@EqualsAndHashCode(callSuper = true)
@RegisterForReflection
@SuperBuilder
@NoArgsConstructor
@Entity
public class ImportProfile extends BaseModel {

    /**
     * Target entity class name (fully qualified).
     * If null, profile can be used with any entity type.
     */
    private String targetType;

    /**
     * Column-to-field mappings with transformations.
     * Order matters - defines CSV column order when requestedColumns not specified.
     */
    private List<ColumnMapping> columnMappings;

    /**
     * Global transformations applied to all string fields.
     */
    private GlobalTransformations globalTransformations;

    /**
     * Whether to fail on first transformation error or collect all errors.
     */
    private boolean failFast = false;
}
```

```

/**
 * Optional description of what this profile is for.
 */
private String description;

/**
 * Optional column name containing per-row intent.
 * If specified, the CSV can include a column with values: INSERT, UPDATE, SKIP.
 * If null, intent is determined automatically (UPSERT behavior).
 *
 * Note: Only INSERT, UPDATE, and SKIP are supported. MERGE and DELETE
 * are intentionally not supported for safety reasons.
 */
private String intentColumn;

/**
 * Default intent when intentColumn is specified but a row has empty/invalid
value.
 * One of: INSERT, UPDATE, UPSERT (auto-detect), SKIP.
 * Default: UPSERT (auto-detect based on refName existence).
 */
private String defaultIntent = "UPSERT";

@Override
public String bmFunctionalArea() {
    return "IMPORTS";
}

@Override
public String bmFunctionalDomain() {
    return "IMPORTS";
}
}

```

3.2.2. 2. ColumnMapping Configuration

```

package com.e2eq.framework.model.persistent.imports;

import lombok.Data;
import lombok.NoArgsConstructor;
import lombok.AllArgsConstructor;
import lombok.Builder;

import java.util.Map;

/**
 * Configuration for mapping and transforming a single CSV column.
 */
@Data

```

```

@NoArgsConstructor
@AllArgsConstructor
@Builder
public class ColumnMapping {

    /**
     * CSV column index (0-based) or column name if header present.
     */
    private String sourceColumn;

    /**
     * Target field name on the entity.
     * Supports nested paths (e.g., "address.city") and array notation (e.g.,
"tags[0]").
     */
    private String targetField;

    /**
     * Static value mappings: CSV value → target value.
     * Applied before type conversion.
     * Example: {"Y": "true", "N": "false"}
     */
    private Map<String, String> valueMappings;

    /**
     * Whether value mapping comparison is case-sensitive.
     * Default: false (case-insensitive)
     */
    @Builder.Default
    private boolean valueMappingCaseSensitive = false;

    /**
     * Behavior when CSV value not found in valueMappings.
     * Default: PASSTHROUGH
     */
    @Builder.Default
    private UnmappedValueBehavior unmappedValueBehavior = UnmappedValueBehavior
.PASSTHROUGH;

    /**
     * Regex pattern for replacement.
     */
    private String regexPattern;

    /**
     * Replacement string for regex matches.
     * Supports capture group references ($1, $2, etc.)
     */
    private String regexReplacement;

    /**

```

```

    * Default value if CSV cell is empty/null.
    * Applied after all other transformations.
    */
    private String defaultValue;

    /**
     * Trim leading/trailing whitespace.
     * Default: true
     */
    @Builder.Default
    private boolean trim = true;

    /**
     * Convert empty string to null.
     * Default: true
     */
    @Builder.Default
    private boolean emptyToNull = true;

    /**
     * Case transformation: UPPER, LOWER, TITLE, NONE.
     * Default: NONE
     */
    @Builder.Default
    private CaseTransform caseTransform = CaseTransform.NONE;

    /**
     * Lookup configuration for resolving references from other collections.
     */
    private LookupConfig lookup;

    /**
     * Date/time format pattern for parsing date strings.
     * Example: "yyyy-MM-dd", "MM/dd/yyyy HH:mm:ss"
     */
    private String dateFormat;

    /**
     * Locale for date/number parsing (e.g., "en-US", "de-DE").
     */
    private String locale;
}

```

3.2.3. 3. Supporting Enums and Classes

```

package com.e2eq.framework.model.persistent.imports;

/**
 * Behavior when a CSV value is not found in the value mappings.

```

```

*/
public enum UnmappedValueBehavior {
    /** Keep the original CSV value unchanged */
    PASSTHROUGH,
    /** Convert to null */
    NULL,
    /** Treat as an error */
    FAIL
}

/**
 * String case transformation options.
 */
public enum CaseTransform {
    NONE,
    UPPER,
    LOWER,
    TITLE
}

```

3.2.4. 4. LookupConfig for Reference Resolution

Supports looking up values from other collections (e.g., map customer name to customer ID).

```

package com.e2eq.framework.model.persistent.imports;

import lombok.Data;
import lombok.NoArgsConstructor;
import lombok.AllArgsConstructor;
import lombok.Builder;

/**
 * Configuration for looking up values from another collection.
 * Useful for resolving foreign key references during import.
 *
 * Example: Map customer name from CSV to customer refName in database.
 */
@Data
@NoArgsConstructor
@AllArgsConstructor
@Builder
public class LookupConfig {

    /**
     * Collection/entity class name to lookup from.
     * Can be simple name (e.g., "Customer") or fully qualified.
     */
    private String lookupCollection;

    /**

```

```

    * Field in lookup collection to match against CSV value.
    * Example: "displayName", "email", "externalId"
    */
    private String lookupMatchField;

    /**
     * Field in lookup collection to return as the mapped value.
     * Example: "refName", "id"
     */
    private String lookupReturnField;

    /**
     * Behavior when lookup fails to find a match.
     * Default: FAIL
     */
    @Builder.Default
    private LookupFailBehavior onNotFound = LookupFailBehavior.FAIL;

    /**
     * Whether to cache lookup results for performance.
     * Recommended for large imports with repeated values.
     * Default: true
     */
    @Builder.Default
    private boolean cacheLookups = true;

    /**
     * Optional filter to apply to lookup query.
     * Uses standard Quantum query syntax.
     * Example: "status:ACTIVE"
     */
    private String lookupFilter;
}

/**
 * Behavior when lookup fails to find a matching record.
 */
public enum LookupFailBehavior {
    /** Mark row as error */
    FAIL,
    /** Set field to null */
    NULL,
    /** Keep original CSV value unchanged */
    PASSTHROUGH
}

```

3.2.5. 5. Header-Based Field Metadata

CSV headers can include optional suffix modifiers to specify field behavior. This feature is **opt-in** via the `enableHeaderModifiers` flag.

Header Modifier Syntax

```
fieldName[modifier]
```

Where modifier is one of:

- * = Required (fail if empty/null)
- ? = Optional (allow empty/null, skip validation)
- ~ = Calculated (compute if not provided)
- # = Key field (used for intent detection instead of refName)

Examples

```
refName*,displayName*,description?,price*,sku#,createdAt~,modifiedAt~
SKU-001,Widget,A great widget,19.99,SKU-001,,
SKU-002,Gadget,,29.99,SKU-002,,
```

In this example: - **refName*** and **displayName*** are required - import fails if empty - **description?** is optional - empty values are accepted without validation error - **price*** is required - **sku#** is the key field - used instead of **refName** for INSERT vs UPDATE detection - **createdAt~** and **modifiedAt~** are calculated - if empty, system generates values

HeaderModifier Enum

```
package com.e2eq.framework.util.csv;

/**
 * Modifiers that can be applied to CSV header columns via suffix notation.
 * Feature is opt-in via ImportProfile.enableHeaderModifiers or REST parameter.
 */
public enum HeaderModifier {
    /**
     * No modifier - field uses default behavior from entity validation.
     */
    NONE(""),

    /**
     * Required field - import fails if value is empty/null.
     * Checked BEFORE Jakarta validation, providing earlier feedback.
     * Suffix: *
     */
    REQUIRED("*"),

    /**
     * Optional field - empty/null values are allowed.
     * Skips @NotNull/@NotEmpty validation for this field.
     * Suffix: ?
     */
    OPTIONAL("?"),
```

```

/**
 * Calculated field - if empty, value is computed by a FieldCalculator.
 * Common uses: timestamps, UUIDs, derived values.
 * Suffix: ~
 */
CALCULATED("~"),

/**
 * Key field - used for INSERT vs UPDATE detection instead of refName.
 * Useful when importing data where refName differs from natural key.
 * Only one key field allowed per import.
 * Suffix: #
 */
KEY("#");

private final String suffix;

HeaderModifier(String suffix) {
    this.suffix = suffix;
}

public String getSuffix() {
    return suffix;
}

/**
 * Parse modifier from header column name.
 * @param header The raw header (e.g., "displayName*", "price?")
 * @return The modifier, or NONE if no recognized suffix
 */
public static HeaderModifier fromHeader(String header) {
    if (header == null || header.isEmpty()) return NONE;
    char last = header.charAt(header.length() - 1);
    return switch (last) {
        case '*' -> REQUIRED;
        case '?' -> OPTIONAL;
        case '~' -> CALCULATED;
        case '#' -> KEY;
        default -> NONE;
    };
}

/**
 * Strip modifier suffix from header to get field name.
 * @param header The raw header (e.g., "displayName*")
 * @return The field name without suffix (e.g., "displayName")
 */
public static String stripModifier(String header) {
    if (header == null || header.isEmpty()) return header;
    char last = header.charAt(header.length() - 1);

```

```

        if (last == '*' || last == '?' || last == '~' || last == '#') {
            return header.substring(0, header.length() - 1);
        }
        return header;
    }
}

```

ParsedHeader Class

```

package com.e2eq.framework.util.csv;

import lombok.Value;

/**
 * Represents a parsed CSV header with field name and modifier.
 */
@Value
public class ParsedHeader {
    String fieldName;
    HeaderModifier modifier;

    /**
     * Parse a raw header string into field name and modifier.
     */
    public static ParsedHeader parse(String rawHeader) {
        HeaderModifier modifier = HeaderModifier.fromHeader(rawHeader);
        String fieldName = HeaderModifier.stripModifier(rawHeader);
        return new ParsedHeader(fieldName, modifier);
    }

    /**
     * @return true if this field is marked as required (*)
     */
    public boolean isRequired() {
        return modifier == HeaderModifier.REQUIRED;
    }

    /**
     * @return true if this field is marked as optional
     */
    public boolean isOptional() {
        return modifier == HeaderModifier.OPTIONAL;
    }

    /**
     * @return true if this field should be calculated when empty
     */
    public boolean isCalculated() {
        return modifier == HeaderModifier.CALCULATED;
    }
}

```

```

/**
 * @return true if this field is the key for intent detection (#)
 */
public boolean isKey() {
    return modifier == HeaderModifier.KEY;
}
}

```

FieldCalculator SPI

For calculated fields (~ modifier), a pluggable calculator interface:

```

package com.e2eq.framework.util.csv;

import com.e2eq.framework.model.persistent.base.UnversionedBaseModel;

/**
 * SPI for calculating field values when marked with ~ modifier and value is empty.
 * Implementations are discovered via CDI.
 */
public interface FieldCalculator {

    /**
     * @return The field name this calculator handles (e.g., "createdAt", "uuid")
     */
    String getFieldName();

    /**
     * @return Entity types this calculator applies to, or empty for all types
     */
    default Class<?>[] getTargetTypes() {
        return new Class<?>[0]; // All types
    }

    /**
     * @return Priority (lower = earlier). Default calculators run at 1000.
     */
    default int getPriority() {
        return 1000;
    }

    /**
     * Calculate the value for a field.
     *
     * @param bean The entity being imported
     * @param fieldName The field to calculate
     * @param context Import context
     * @return The calculated value
     */
}

```

```
    Object calculate(UnversionedBaseModel bean, String fieldName, ImportRowContext context);
}
```

Built-in Field Calculators

```
/**
 * Calculates timestamp fields (createdAt, modifiedAt, updatedAt).
 */
@ApplicationScoped
public class TimestampFieldCalculator implements FieldCalculator {

    private static final Set<String> TIMESTAMP_FIELDS = Set.of(
        "createdAt", "modifiedAt", "updatedAt", "createdDate", "modifiedDate"
    );

    @Override
    public String getFieldName() {
        return null; // Handles multiple fields
    }

    public boolean handles(String fieldName) {
        return TIMESTAMP_FIELDS.contains(fieldName);
    }

    @Override
    public Object calculate(UnversionedBaseModel bean, String fieldName,
        ImportRowContext context) {
        return Instant.now();
    }
}

/**
 * Calculates UUID fields.
 */
@ApplicationScoped
public class UUIDFieldCalculator implements FieldCalculator {

    @Override
    public String getFieldName() {
        return "uuid";
    }

    @Override
    public Object calculate(UnversionedBaseModel bean, String fieldName,
        ImportRowContext context) {
        return UUID.randomUUID().toString();
    }
}
```

```

/**
 * Calculates refName from other fields if not provided.
 */
@ApplicationScoped
public class RefNameCalculator implements FieldCalculator {

    @Override
    public String getFieldName() {
        return "refName";
    }

    @Override
    public int getPriority() {
        return 500; // Run early
    }

    @Override
    public Object calculate(UnversionedBaseModel bean, String fieldName,
ImportRowContext context) {
        // Try to derive from key field or generate UUID
        if (bean.getDisplayName() != null && !bean.getDisplayName().isBlank()) {
            return slugify(bean.getDisplayName());
        }
        return UUID.randomUUID().toString();
    }

    private String slugify(String input) {
        return input.toLowerCase()
            .replaceAll("[^a-z0-9]+", "-")
            .replaceAll("^-|-$", "");
    }
}

```

ImportProfile Configuration

Add to ImportProfile:

```

/**
 * Enable header modifier parsing (!, ?, ~, *).
 * When false (default), headers are treated as plain field names.
 * This maintains backward compatibility.
 */
private boolean enableHeaderModifiers = false;

/**
 * Custom field calculators defined in the profile.
 * These supplement CDI-discovered calculators.
 */
private List<InlineFieldCalculator> inlineCalculators;

```

```

/**
 * Inline field calculator defined within an ImportProfile.
 * Supports simple expression-based calculations.
 */
@Data
public class InlineFieldCalculator {
    /**
     * Field name to calculate.
     */
    private String fieldName;

    /**
     * Calculation type.
     */
    private CalculationType type;

    /**
     * Static value (for STATIC type).
     */
    private String staticValue;

    /**
     * Source field (for COPY type).
     */
    private String sourceField;

    /**
     * Template with ${fieldName} placeholders (for TEMPLATE type).
     */
    private String template;

    public enum CalculationType {
        /** Use current timestamp */
        TIMESTAMP,
        /** Generate UUID */
        UUID,
        /** Use static value */
        STATIC,
        /** Copy from another field */
        COPY,
        /** Template with field placeholders */
        TEMPLATE
    }
}

```

REST API Parameter

```

@Parameter(description = "Enable header modifier parsing (*, ?, ~, #). " +
    "When true, headers like 'name*' are parsed as required

```

```
fields.")
@QueryParam("enableHeaderModifiers") @DefaultValue("false") boolean
enableHeaderModifiers
```

Processing Logic

```
/**
 * Process headers and extract field metadata.
 */
private List<ParsedHeader> parseHeaders(
    String[] rawHeaders,
    boolean enableModifiers) {

    List<ParsedHeader> parsed = new ArrayList<>(rawHeaders.length);
    int keyFieldCount = 0;

    for (String raw : rawHeaders) {
        if (enableModifiers) {
            ParsedHeader header = ParsedHeader.parse(raw);
            if (header.isKey()) {
                keyFieldCount++;
                if (keyFieldCount > 1) {
                    throw new ImportException(
                        "Multiple key fields (#) specified. Only one key field is
allowed.");
                }
            }
            parsed.add(header);
        } else {
            // No modifier parsing - treat as plain field name
            parsed.add(new ParsedHeader(raw, HeaderModifier.NONE));
        }
    }

    return parsed;
}

/**
 * Validate required fields after parsing, before Jakarta validation.
 */
private void validateRequiredFields(
    Object bean,
    List<ParsedHeader> headers,
    Map<String, Object> values,
    ImportRowResult<?> rowResult) {

    for (ParsedHeader header : headers) {
        if (header.isRequired()) {
            Object value = values.get(header.getFieldName());
            if (value == null || (value instanceof String s && s.isBlank())) {
```

```

        rowResult.addError(new FieldError(
            header.getFieldName(),
            "Field is required (marked with *) but value is empty",
            FieldErrorCode.VALIDATION
        ));
    }
}

/**
 * Apply calculated fields.
 */
private void applyCalculatedFields(
    UnversionedBaseModel bean,
    List<ParsedHeader> headers,
    ImportRowContext context) {

    for (ParsedHeader header : headers) {
        if (header.isCalculated()) {
            Object currentValue = getFieldValue(bean, header.getFieldName());
            if (currentValue == null || (currentValue instanceof String s && s.
isBlank())) {
                Object calculated = calculateField(bean, header.getFieldName(),
context);
                if (calculated != null) {
                    setFieldValue(bean, header.getFieldName(), calculated);
                }
            }
        }
    }
}

/**
 * Find key field for intent detection.
 */
private String findKeyField(List<ParsedHeader> headers) {
    for (ParsedHeader header : headers) {
        if (header.isKey()) {
            return header.getFieldName();
        }
    }
    return "refName"; // Default
}

```

Example: Complete CSV with All Modifiers

```

_action,sku#,displayName*,description?,price*,category*,createdAt~,uuid~
INSERT,NEW-001,New Widget,A great new widget,19.99,Electronics,,
UPDATE,EXIST-001,Updated Name,,29.99,Home,,

```

```
UPSERT,AUTO-001,Auto Detect,Optional desc,39.99,Tools,,
SKIP,IGNORE-001,Ignored Row,,,,,
```

Header interpretation:

Header	Modifier	Behavior
<code>_action</code>	none	Intent column (configured separately)
<code>sku#</code>	KEY	Use <code>sku</code> instead of <code>refName</code> for INSERT/UPDATE detection
<code>displayName*</code>	REQUIRED	Fail import if empty
<code>description?</code>	OPTIONAL	Allow empty, skip @NotNull validation
<code>price*</code>	REQUIRED	Fail import if empty
<code>category*</code>	REQUIRED	Fail import if empty
<code>createdAt~</code>	CALCULATED	Generate timestamp if empty
<code>uuid~</code>	CALCULATED	Generate UUID if empty

Backward Compatibility

- **Default behavior unchanged:** `enableHeaderModifiers=false` by default
- **Opt-in only:** Must explicitly enable via profile or REST parameter
- **Graceful handling:** If disabled, characters like `*?~#` in field names are treated literally
- **Escape mechanism:** To use literal `in a field name when modifiers are enabled`, use `\\`

3.2.6. 6. Intent Configuration

The existing `Intent` enum (INSERT, UPDATE, SKIP) can optionally be specified per-row via a designated CSV column.

```
package com.e2eq.framework.util.csv;

/**
 * Supported intent values for CSV import.
 * These match the existing CSVImportHelper.Intent enum.
 *
 * NOTE: MERGE and DELETE are intentionally NOT supported to prevent
 * accidental data loss or complex partial updates via CSV.
 */
public enum ImportIntent {
    /**
     * Insert a new record. Fails if refName already exists.
     */
    INSERT,

    /**
```

```

    * Update an existing record. Fails if refName does not exist.
    */
    UPDATE,

    /**
     * Auto-detect: INSERT if new, UPDATE if exists.
     * This is the current default behavior.
     */
    UPSERT,

    /**
     * Skip this row - do not process.
     * Useful for conditional imports or marking rows to ignore.
     */
    SKIP
}

```

Why MERGE and DELETE are not supported:

- **DELETE:** Allowing bulk deletes via CSV is dangerous. A malformed file or user error could delete large amounts of data. Deletes should be explicit via the REST DELETE endpoint or purpose-built batch operations with additional safeguards.
- **MERGE:** Partial/merge updates are complex and error-prone. It's unclear which fields should be preserved vs. overwritten when the CSV value is empty. The current full-replacement semantic is predictable and explicit.

3.2.7. 7. GlobalTransformations

```

package com.e2eq.framework.model.persistent.imports;

import lombok.Data;
import lombok.NoArgsConstructor;
import lombok.AllArgsConstructor;
import lombok.Builder;

/**
 * Global transformations applied to all string fields during import.
 * These are applied before column-specific transformations.
 */
@Data
@NoArgsConstructor
@AllArgsConstructor
@Builder
public class GlobalTransformations {

    /**
     * Trim leading/trailing whitespace from all string fields.
     * Default: true
     */
}

```

```

@Builder.Default
private boolean trimStrings = true;

/**
 * Convert empty strings to null.
 * Default: true
 */
@Builder.Default
private boolean emptyStringsToNull = true;

/**
 * Unicode normalization form: NFC, NFD, NFKC, NFKD, or null for none.
 */
private String unicodeNormalization;

/**
 * Remove ASCII control characters (0x00-0x1F except tab/newline).
 * Default: false
 */
@Builder.Default
private boolean removeControlChars = false;

/**
 * Maximum string length. Strings exceeding this are truncated.
 * Null means no limit.
 */
private Integer maxStringLength;

/**
 * Replace multiple consecutive whitespace with single space.
 * Default: false
 */
@Builder.Default
private boolean normalizeWhitespace = false;
}

```

3.2.8. 8. LookupService Interface

```

package com.e2eq.framework.util.csv;

import com.e2eq.framework.model.persistent.imports.LookupConfig;
import java.util.List;
import java.util.Map;
import java.util.Optional;

/**
 * Service for resolving lookup references during CSV import.
 */
public interface LookupService {

```

```

/**
 * Look up a single value.
 *
 * @param config The lookup configuration
 * @param matchValue The value to match against lookupMatchField
 * @return The resolved value, or empty if not found
 */
Optional<String> lookup(LookupConfig config, String matchValue);

/**
 * Batch lookup for performance optimization.
 * Implementations should use IN queries where possible.
 *
 * @param config The lookup configuration
 * @param matchValues The values to match
 * @return Map of matchValue → resolvedValue (missing keys = not found)
 */
Map<String, String> batchLookup(LookupConfig config, List<String> matchValues);

/**
 * Clear any cached lookup results.
 */
void clearCache();
}

```

3.2.9. 9. PreValidationTransformer Interface

Programmatic hook for complex transformations that can't be expressed declaratively.

```

package com.e2eq.framework.util.csv;

import com.e2eq.framework.model.persistent.base.UnversionedBaseModel;

/**
 * SPI for transforming parsed CSV beans before validation.
 * Implementations are discovered via CDI and applied in priority order.
 *
 * Use cases:
 * - Computing derived fields
 * - Cross-field validation/transformation
 * - Complex business logic that can't be expressed declaratively
 * - Conditional transformations based on other field values
 */
public interface PreValidationTransformer<T extends UnversionedBaseModel> {

    /**
     * @return The entity class this transformer handles
     */
    Class<T> getTargetType();
}

```

```

/**
 * @return Priority (lower = earlier). Default transformers run at 1000.
 */
default int getPriority() {
    return 1000;
}

/**
 * Transform the bean after CSV parsing but before validation.
 *
 * @param bean The parsed bean instance (already has cell processor
transformations applied)
 * @param context Import context with access to row data, profile, lookup service,
etc.
 * @return The transformed bean (may be same instance modified in place, or a new
instance)
 * @throws TransformationException if transformation fails (row will be marked as
error)
 */
T transform(T bean, ImportRowContext context) throws TransformationException;
}

```

3.2.10. 10. ImportRowContext

```

package com.e2eq.framework.util.csv;

import com.e2eq.framework.model.persistent.imports.ImportProfile;
import lombok.Builder;
import lombok.Data;

import java.util.List;
import java.util.Map;

/**
 * Context information available to PreValidationTransformer implementations.
 */
@Data
@Builder
public class ImportRowContext {

    /**
     * Current row number (1-based, after header).
     */
    private int rowNumber;

    /**
     * Raw CSV line as string.
     */
}

```

```

private String rawLine;

/**
 * Parsed cell values by column index (0-based).
 * These are the raw string values before any transformation.
 */
private List<String> rawValues;

/**
 * Parsed cell values by column name (if header was present).
 * These are the raw string values before any transformation.
 */
private Map<String, String> rawValuesByName;

/**
 * The import profile being used (may be null for profile-less imports).
 */
private ImportProfile profile;

/**
 * The import session ID (if using session-based flow).
 */
private String sessionId;

/**
 * Access to lookup service for additional reference resolution.
 */
private LookupService lookupService;
}

```

3.2.11. 11. TransformationException

```

package com.e2eq.framework.util.csv;

/**
 * Exception thrown when a transformation fails during CSV import.
 * Results in the row being marked as an error.
 */
public class TransformationException extends RuntimeException {

    private final String field;

    public TransformationException(String message) {
        super(message);
        this.field = null;
    }

    public TransformationException(String field, String message) {
        super(message);
    }
}

```

```

        this.field = field;
    }

    public TransformationException(String message, Throwable cause) {
        super(message, cause);
        this.field = null;
    }

    public TransformationException(String field, String message, Throwable cause) {
        super(message, cause);
        this.field = field;
    }

    /**
     * @return The field that caused the error, or null if not field-specific
     */
    public String getField() {
        return field;
    }
}

```

3.3. Enhanced Cell Processors

3.3.1. ValueMapProcessor

```

package com.e2eq.framework.util.csv.processors;

import com.e2eq.framework.model.persistent.imports.UnmappedValueBehavior;
import org.supercsv.cellprocessor.CellProcessorAdaptor;
import org.supercsv.cellprocessor.ift.CellProcessor;
import org.supercsv.exception.SuperCsvCellProcessorException;
import org.supercsv.util.CsvContext;

import java.util.HashMap;
import java.util.Map;

/**
 * Cell processor that maps values using a lookup table.
 */
public class ValueMapProcessor extends CellProcessorAdaptor {

    private final Map<String, String> mappings;
    private final Map<String, String> lowerCaseMappings;
    private final boolean caseSensitive;
    private final UnmappedValueBehavior unmappedBehavior;

    public ValueMapProcessor(
        Map<String, String> mappings,
        boolean caseSensitive,

```

```

        UnmappedValueBehavior unmappedBehavior,
        CellProcessor next) {
    super(next);
    this.mappings = mappings;
    this.caseSensitive = caseSensitive;
    this.unmappedBehavior = unmappedBehavior;

    // Pre-compute lowercase keys for case-insensitive matching
    if (!caseSensitive) {
        this.lowerCaseMappings = new HashMap<>();
        mappings.forEach((k, v) -> lowerCaseMappings.put(k.toLowerCase(), v));
    } else {
        this.lowerCaseMappings = null;
    }
}

@Override
public Object execute(Object value, CsvContext ctx) {
    if (value == null) {
        return next.execute(null, ctx);
    }

    String stringValue = value.toString();
    String mapped;

    if (caseSensitive) {
        mapped = mappings.get(stringValue);
    } else {
        mapped = lowerCaseMappings.get(stringValue.toLowerCase());
    }

    if (mapped == null) {
        switch (unmappedBehavior) {
            case PASSTHROUGH:
                return next.execute(value, ctx);
            case NULL:
                return next.execute(null, ctx);
            case FAIL:
                throw new SuperCsvCellProcessorException(
                    String.format("No mapping found for value '%s'. Valid values:
%s",
                                stringValue, mappings.keySet()),
                    ctx, this);
            default:
                return next.execute(value, ctx);
        }
    }

    return next.execute(mapped, ctx);
}

```

```
}
```

3.3.2. RegexReplaceProcessor

```
package com.e2eq.framework.util.csv.processors;

import org.supercsv.cellprocessor.CellProcessorAdaptor;
import org.supercsv.cellprocessor.ift.CellProcessor;
import org.supercsv.util.CsvContext;

import java.util.regex.Pattern;

/**
 * Cell processor that performs regex replacement on string values.
 */
public class RegexReplaceProcessor extends CellProcessorAdaptor {

    private final Pattern pattern;
    private final String replacement;

    public RegexReplaceProcessor(String regex, String replacement, CellProcessor next)
    {
        super(next);
        this.pattern = Pattern.compile(regex);
        this.replacement = replacement != null ? replacement : "";
    }

    @Override
    public Object execute(Object value, CsvContext ctx) {
        if (value == null) {
            return next.execute(null, ctx);
        }

        String result = pattern.matcher(value.toString()).replaceAll(replacement);
        return next.execute(result, ctx);
    }
}
```

3.3.3. CaseTransformProcessor

```
package com.e2eq.framework.util.csv.processors;

import com.e2eq.framework.model.persistent.imports.CaseTransform;
import org.supercsv.cellprocessor.CellProcessorAdaptor;
import org.supercsv.cellprocessor.ift.CellProcessor;
import org.supercsv.util.CsvContext;

/**
```

```

* Cell processor that transforms string case.
*/
public class CaseTransformProcessor extends CellProcessorAdaptor {

    private final CaseTransform transform;

    public CaseTransformProcessor(CaseTransform transform, CellProcessor next) {
        super(next);
        this.transform = transform;
    }

    @Override
    public Object execute(Object value, CsvContext ctx) {
        if (value == null) {
            return next.execute(null, ctx);
        }

        String s = value.toString();
        String result = switch (transform) {
            case UPPER -> s.toUpperCase();
            case LOWER -> s.toLowerCase();
            case TITLE -> toTitleCase(s);
            case NONE -> s;
        };

        return next.execute(result, ctx);
    }

    private String toTitleCase(String s) {
        if (s == null || s.isEmpty()) return s;

        StringBuilder result = new StringBuilder(s.length());
        boolean capitalizeNext = true;

        for (char c : s.toCharArray()) {
            if (Character.isWhitespace(c)) {
                capitalizeNext = true;
                result.append(c);
            } else if (capitalizeNext) {
                result.append(Character.toUpperCase(c));
                capitalizeNext = false;
            } else {
                result.append(Character.toLowerCase(c));
            }
        }

        return result.toString();
    }
}

```

3.3.4. LookupProcessor

```
package com.e2eq.framework.util.csv.processors;

import com.e2eq.framework.model.persistent.imports.LookupConfig;
import com.e2eq.framework.model.persistent.imports.LookupFailBehavior;
import com.e2eq.framework.util.csv.LookupService;
import org.supercsv.cellprocessor.CellProcessorAdaptor;
import org.supercsv.cellprocessor.ift.CellProcessor;
import org.supercsv.exception.SuperCsvCellProcessorException;
import org.supercsv.util.CsvContext;

import java.util.Map;
import java.util.Optional;
import java.util.concurrent.ConcurrentHashMap;

/**
 * Cell processor that looks up values from another collection.
 */
public class LookupProcessor extends CellProcessorAdaptor {

    private final LookupService lookupService;
    private final LookupConfig config;
    private final Map<String, Optional<String>> cache;

    public LookupProcessor(
        LookupService lookupService,
        LookupConfig config,
        CellProcessor next) {
        super(next);
        this.lookupService = lookupService;
        this.config = config;
        this.cache = config.isCacheLookups() ? new ConcurrentHashMap<>() : null;
    }

    @Override
    public Object execute(Object value, CsvContext ctx) {
        if (value == null) {
            return next.execute(null, ctx);
        }

        String key = value.toString();
        Optional<String> result;

        if (cache != null) {
            result = cache.computeIfAbsent(key, k -> lookupService.lookup(config, k));
        } else {
            result = lookupService.lookup(config, key);
        }
    }
}
```

```

        if (result.isEmpty()) {
            switch (config.getOnNotFound()) {
                case FAIL:
                    throw new SuperCsvCellProcessorException(
                        String.format("Lookup failed: no %s found with %s = '%s'",
                            config.getLookupCollection(),
                            config.getLookupMatchField(),
                            key),
                        ctx, this);
                case NULL:
                    return next.execute(null, ctx);
                case PASSTHROUGH:
                    return next.execute(value, ctx);
            }
        }

        return next.execute(result.get(), ctx);
    }

    /**
     * Clear the lookup cache. Call between imports if reusing processor.
     */
    public void clearCache() {
        if (cache != null) {
            cache.clear();
        }
    }
}

```

3.4. Enhanced CSVImportHelper

3.4.1. New Method Signatures

```

@ApplicationScoped
public class CSVImportHelper {

    @Inject
    Instance<PreValidationTransformer<?>> transformers;

    @Inject
    ImportProfileRepo importProfileRepo;

    @Inject
    LookupService lookupService;

    // ... existing methods unchanged ...

    /**
     * Import CSV with an ImportProfile for transformations.

```

```

*
* @param repo The repository for the target entity type
* @param inputStream The CSV input stream
* @param profile The import profile with transformation rules
* @param formatOptions CSV format options (separator, quote, charset, etc.)
* @param failedRecordHandler Optional handler for failed records
* @return Import result with counts and error details
*/
public <T extends UnversionedBaseModel> ImportResult<T> importCSV(
    BaseMorphiaRepo<T> repo,
    InputStream inputStream,
    ImportProfile profile,
    CSVFormatOptions formatOptions,
    FailedRecordHandler<T> failedRecordHandler);

/**
 * Import CSV with profile referenced by refName.
 *
 * @param repo The repository for the target entity type
 * @param inputStream The CSV input stream
 * @param profileRefName The refName of an existing ImportProfile
 * @param formatOptions CSV format options
 * @param failedRecordHandler Optional handler for failed records
 * @return Import result with counts and error details
 * @throws ValidationException if profile not found
 */
public <T extends UnversionedBaseModel> ImportResult<T> importCSV(
    BaseMorphiaRepo<T> repo,
    InputStream inputStream,
    String profileRefName,
    CSVFormatOptions formatOptions,
    FailedRecordHandler<T> failedRecordHandler);

/**
 * Analyze CSV with profile for preview (session-based flow).
 *
 * @param repo The repository for the target entity type
 * @param inputStream The CSV input stream
 * @param profile The import profile with transformation rules
 * @param formatOptions CSV format options
 * @return Import result with preview data and session ID
 */
public <T extends UnversionedBaseModel> ImportResult<T> analyzeCSV(
    BaseMorphiaRepo<T> repo,
    InputStream inputStream,
    ImportProfile profile,
    CSVFormatOptions formatOptions) throws IOException;
}

```

3.4.2. CSVFormatOptions (Refactored)

```
package com.e2eq.framework.util.csv;

import lombok.Builder;
import lombok.Data;

import java.nio.charset.Charset;
import java.nio.charset.StandardCharsets;
import java.util.List;

/**
 * CSV format configuration options.
 */
@Data
@Builder
public class CSVFormatOptions {

    @Builder.Default
    private char fieldSeparator = ',';

    @Builder.Default
    private char quoteChar = '"';

    @Builder.Default
    private boolean skipHeaderRow = true;

    @Builder.Default
    private Charset charset = StandardCharsets.UTF_8;

    @Builder.Default
    private boolean hasBOM = false;

    @Builder.Default
    private String quotingStrategy = "QUOTE_WHERE_ESSENTIAL";

    /**
     * Column names/fields to import, in CSV column order.
     * If null and profile has columnMappings, uses those.
     * If both null, uses header row.
     */
    private List<String> requestedColumns;
}
```

3.4.3. Enhanced Processor Chain Building

```
/**
 * Build cell processors from ImportProfile column mappings.
 */
```

```

private CellProcessor[] buildProcessorsFromProfile(
    Class<?> clazz,
    ImportProfile profile,
    ListCellProcessor listProcessor) {

    List<ColumnMapping> mappings = profile.getColumnMappings();
    CellProcessor[] processors = new CellProcessor[mappings.size()];

    for (int i = 0; i < mappings.size(); i++) {
        ColumnMapping mapping = mappings.get(i);
        processors[i] = buildProcessorChain(clazz, mapping, listProcessor);
    }

    return processors;
}

/**
 * Build a processor chain for a single column.
 * Processors are chained innermost-first (type conversion is innermost).
 */
private CellProcessor buildProcessorChain(
    Class<?> clazz,
    ColumnMapping mapping,
    ListCellProcessor listProcessor) {

    String fieldName = mapping.getTargetField();
    Class<?> type = getFieldType(clazz, fieldName);

    // Start with type-specific processor (innermost)
    CellProcessor chain = buildTypeProcessor(type, mapping, listProcessor);

    // Add lookup if configured
    if (mapping.getLookup() != null) {
        chain = new LookupProcessor(lookupService, mapping.getLookup(), chain);
    }

    // Add value mapping if configured
    if (mapping.getValueMappings() != null && !mapping.getValueMappings().isEmpty()) {
        chain = new ValueMapProcessor(
            mapping.getValueMappings(),
            mapping.isValueMappingCaseSensitive(),
            mapping.getUnmappedValueBehavior(),
            chain);
    }

    // Add regex replacement if configured
    if (mapping.getRegexPattern() != null) {
        chain = new RegexReplaceProcessor(
            mapping.getRegexPattern(),
            mapping.getRegexReplacement(),
            chain);
    }
}

```

```

    }

    // Add case transformation
    if (mapping.getCaseTransform() != CaseTransform.NONE) {
        chain = new CaseTransformProcessor(mapping.getCaseTransform(), chain);
    }

    // Add trim (applied early, before other transformations)
    if (mapping.isTrim()) {
        chain = new org.supercsv.cellprocessor.Trim(chain);
    }

    // Add empty-to-null conversion
    if (mapping.isEmptyToNull()) {
        chain = new EmptyToNullProcessor(chain);
    }

    // Add default value handling (applied last, after null conversion)
    if (mapping.getDefaultValue() != null) {
        chain = new org.supercsv.cellprocessor.ConvertNullTo(mapping.getDefaultValue(
), chain);
    }

    // Wrap with Optional for null safety
    return new org.supercsv.cellprocessor.Optional(chain);
}

private CellProcessor buildTypeProcessor(
    Class<?> type,
    ColumnMapping mapping,
    ListCellProcessor listProcessor) {

    if (mapping.getTargetField().contains("[")) {
        return listProcessor;
    }

    // Handle date with custom format
    if (type == java.time.LocalDate.class || type == java.time.LocalDateTime.class
        || type == java.util.Date.class) {
        if (mapping.getDateFormat() != null) {
            return new ParseDateProcessor(mapping.getDateFormat(), mapping.getLocale(
));
        }
    }

    // Standard type processors
    if (type == int.class || type == Integer.class) {
        return new org.supercsv.cellprocessor.ParseInt();
    } else if (type == long.class || type == Long.class) {
        return new org.supercsv.cellprocessor.ParseLong();
    } else if (type == double.class || type == Double.class)

```

```

        || type == float.class || type == Float.class) {
            return new org.supercsv.cellprocessor.ParseDouble();
        } else if (type == java.math.BigDecimal.class) {
            return new org.supercsv.cellprocessor.ParseBigDecimal();
        } else if (type == boolean.class || type == Boolean.class) {
            return new org.supercsv.cellprocessor.ParseBool();
        } else if (type != null && type.isEnum()) {
            return new ParseEnum((Class<? extends Enum<?>>) type);
        }

        // Default: pass through as string
        return new org.supercsv.cellprocessor.constraint.NotNull();
    }
}

```

3.4.4. PreValidationTransformer Integration

```

/**
 * Apply pre-validation transformers to a parsed bean.
 */
@SuppressWarnings("unchecked")
private <T extends UnversionedBaseModel> T applyTransformers(
    T bean,
    ImportRowContext context) throws TransformationException {

    if (transformers == null || transformers.isUnsatisfied()) {
        return bean;
    }

    // Collect and sort transformers by priority
    List<PreValidationTransformer<T>> applicable = new ArrayList<>();
    for (PreValidationTransformer<?> transformer : transformers) {
        if (transformer.getTargetType().isAssignableFrom(bean.getClass())) {
            applicable.add((PreValidationTransformer<T>) transformer);
        }
    }

    applicable.sort(Comparator.comparingInt(PreValidationTransformer::getPriority));

    // Apply transformers in order
    T result = bean;
    for (PreValidationTransformer<T> transformer : applicable) {
        result = transformer.transform(result, context);
    }

    return result;
}

```

3.4.5. Intent Resolution

When an `intentColumn` is configured, the import process reads the intent from each row and validates it.

```
/**
 * Resolve the intent for a row from CSV data or profile defaults.
 *
 * @param profile The import profile (may have intentColumn configured)
 * @param rowValues The parsed CSV row values by column name
 * @param existsInDb Whether the record already exists (for UPSERT resolution)
 * @return The resolved intent for this row
 * @throws ImportException if an invalid intent value is specified
 */
private ImportIntent resolveIntent(
    ImportProfile profile,
    Map<String, String> rowValues,
    boolean existsInDb) {

    // If no intent column configured, use default behavior
    if (profile == null || profile.getIntentColumn() == null) {
        return existsInDb ? ImportIntent.UPDATE : ImportIntent.INSERT;
    }

    // Read intent from CSV column
    String intentValue = rowValues.get(profile.getIntentColumn());

    if (intentValue != null && !intentValue.isBlank()) {
        String normalized = intentValue.toUpperCase().trim();

        // Validate against allowed intents (explicitly reject MERGE/DELETE)
        switch (normalized) {
            case "INSERT":
                return ImportIntent.INSERT;
            case "UPDATE":
                return ImportIntent.UPDATE;
            case "SKIP":
                return ImportIntent.SKIP;
            case "UPSERT":
                return existsInDb ? ImportIntent.UPDATE : ImportIntent.INSERT;
            case "MERGE":
            case "DELETE":
                throw new ImportException(
                    "Intent '" + normalized + "' is not supported. " +
                    "Only INSERT, UPDATE, UPSERT, and SKIP are allowed.");
            default:
                throw new ImportException(
                    "Invalid intent value: '" + intentValue + "'. " +
                    "Valid values are: INSERT, UPDATE, UPSERT, SKIP.");
        }
    }
}
```

```

}

// Fall back to profile default
String defaultIntent = profile.getDefaultIntent();
if (defaultIntent == null || defaultIntent.isBlank()) {
    defaultIntent = "UPSERT";
}

return switch (defaultIntent.toUpperCase()) {
    case "INSERT" -> ImportIntent.INSERT;
    case "UPDATE" -> ImportIntent.UPDATE;
    case "SKIP" -> ImportIntent.SKIP;
    default -> existsInDb ? ImportIntent.UPDATE : ImportIntent.INSERT; // UPSERT
};
}

```

3.4.6. Intent Execution

```

/**
 * Execute the save operation based on the resolved intent.
 */
private <T extends UnversionedBaseModel> void executeWithIntent(
    BaseMorphiaRepo<T> repo,
    T bean,
    ImportIntent intent,
    ImportRowResult<T> rowResult) {

    switch (intent) {
        case INSERT -> {
            // Verify record doesn't exist
            Optional<T> existing = repo.findByRefName(bean.getRefName());
            if (existing.isPresent()) {
                throw new ImportException(
                    "Cannot INSERT: record with refName '" + bean.getRefName() +
                    "' already exists. Use UPDATE or UPSERT intent.");
            }
            repo.save(bean);
            rowResult.setIntent(Intent.INSERT);
        }

        case UPDATE -> {
            // Verify record exists and get its ID
            Optional<T> existing = repo.findByRefName(bean.getRefName());
            if (existing.isEmpty()) {
                throw new ImportException(
                    "Cannot UPDATE: record with refName '" + bean.getRefName() +
                    "' does not exist. Use INSERT or UPSERT intent.");
            }
            // Copy ID to ensure update, not insert

```

```

        bean.setId(existing.get().getId());
        repo.save(bean);
        rowResult.setIntent(Intent.UPDATE);
    }

    case SKIP -> {
        // Do nothing - row is intentionally skipped
        rowResult.setIntent(Intent.SKIP);
    }

    // UPSERT is resolved to INSERT or UPDATE before this method is called
}
}

```

3.5. REST API Enhancements

3.5.1. Updated Endpoints

```

// In BaseResource.java

@POST
@Path("csv")
@Consumes(MediaType.MULTIPART_FORM_DATA)
@Produces(MediaType.APPLICATION_JSON)
@Operation(summary = "Import a list of entities from a CSV file")
public Response importCSVList(
    @Context UriInfo info,
    @BeanParam FileUpload fileUpload,
    // Existing parameters
    @QueryParam("fieldSeparator") @DefaultValue(",") String fieldSeparator,
    @QueryParam("quoteChar") @DefaultValue("\"") String quoteChar,
    @QueryParam("skipHeaderRow") @DefaultValue("true") boolean skipHeaderRow,
    @QueryParam("requestedColumns") String requestedColumns,
    @QueryParam("charsetEncoding") @DefaultValue("UTF-8-without-BOM") String
charsetEncoding,
    @QueryParam("quotingStrategy") @DefaultValue("QUOTE_WHERE_ESSENTIAL") String
quotingStrategy,
    // NEW parameters
    @Parameter(description = "RefName of ImportProfile to use for value mapping
and transformations")
    @QueryParam("profileRefName") String profileRefName,
    @Parameter(description = "CSV column name containing per-row intent (INSERT,
UPDATE, SKIP). " +
                                "If not specified, uses UPSERT behavior (auto-
detect).")
    @QueryParam("intentColumn") String intentColumn,
    @Parameter(description = "Default intent when intentColumn value is empty. " +
                                "One of: INSERT, UPDATE, UPSERT, SKIP. Default:
UPSERT")

```

```

        @QueryParam("defaultIntent") @DefaultValue("UPSERT") String defaultIntent,
        @Parameter(description = "Enable header modifier parsing. When true, headers
like " +
                                "'name*' (required), 'desc?' (optional), 'createdAt~'
(calculated), " +
                                "'sku#' (key field) are parsed for field metadata.")
        @QueryParam("enableHeaderModifiers") @DefaultValue("false") boolean
enableHeaderModifiers
    ) {
        // Implementation uses profile if provided, or parameters directly
    }

@POST
@Path("csv/session")
@Consumes(MediaType.MULTIPART_FORM_DATA)
@Produces(MediaType.APPLICATION_JSON)
@Operation(summary = "Create an import session with preview")
public Response createCsvImportSession(
    // ... same parameters including profileRefName ...
) {
    // Implementation
}

```

3.5.2. ImportProfile Resource

```

package com.e2eq.framework.rest.resources;

import com.e2eq.framework.model.persistent.imports.ImportProfile;
import com.e2eq.framework.model.persistent.morphia.ImportProfileRepo;
import jakarta.ws.rs.Path;

/**
 * REST resource for managing ImportProfiles.
 */
@Path("/integration/import-profiles")
public class ImportProfileResource extends BaseResource<ImportProfile,
ImportProfileRepo> {
    // Standard CRUD endpoints inherited from BaseResource
}

```

3.6. Example Usage

3.6.1. Example ImportProfile (JSON)

```

{
  "refName": "product-import-v1",
  "displayName": "Product Import Profile",

```

```

"description": "Maps legacy product CSV format to Product entity",
"targetType": "com.example.Product",
"columnMappings": [
  {
    "sourceColumn": "SKU",
    "targetField": "refName",
    "trim": true,
    "caseTransform": "UPPER"
  },
  {
    "sourceColumn": "Product Name",
    "targetField": "displayName",
    "trim": true
  },
  {
    "sourceColumn": "Status",
    "targetField": "status",
    "valueMappings": {
      "A": "ACTIVE",
      "I": "INACTIVE",
      "D": "DISCONTINUED",
      "P": "PENDING"
    },
    "unmappedValueBehavior": "FAIL"
  },
  {
    "sourceColumn": "Active",
    "targetField": "isActive",
    "valueMappings": {
      "Y": "true",
      "N": "false",
      "YES": "true",
      "NO": "false",
      "1": "true",
      "0": "false"
    },
    "valueMappingCaseSensitive": false
  },
  {
    "sourceColumn": "Category",
    "targetField": "categoryRefName",
    "lookup": {
      "lookupCollection": "Category",
      "lookupMatchField": "displayName",
      "lookupReturnField": "refName",
      "onNotFound": "FAIL",
      "cacheLookups": true
    }
  },
  {
    "sourceColumn": "Price",

```

```

    "targetField": "price",
    "regexPattern": "[$,]",
    "regexReplacement": "",
    "trim": true
  },
  {
    "sourceColumn": "Tags",
    "targetField": "tags",
    "regexPattern": "\\s*[;,]\\s*",
    "regexReplacement": ","
  }
],
"globalTransformations": {
  "trimStrings": true,
  "emptyStringsToNull": true,
  "normalizeWhitespace": true
},
"intentColumn": "_action",
"defaultIntent": "UPSERT"
}

```

3.6.2. Example CSV (without intent column)

```

SKU,Product Name,Status,Active,Category,Price,Tags
sku-001 ,Widget Pro,A,Y,Electronics,$19.99,new; featured
SKU-002,Gadget Plus,I,N,Home & Garden,$29.99,sale, clearance
sku-003,Tool Kit,D,0,Tools,$49.99,

```

3.6.3. Example CSV (with intent column)

When you need explicit control over whether each row is inserted, updated, or skipped:

```

_action,SKU,Product Name,Status,Active,Category,Price,Tags
INSERT,SKU-NEW-001,Brand New Widget,A,Y,Electronics,$19.99,new
UPDATE,SKU-002,Updated Gadget Name,A,Y,Home & Garden,$34.99,updated
SKIP,SKU-003,Ignore This Row,I,N,Tools,$0.00,
UPSERT,SKU-004,Auto Detect Mode,A,Y,Electronics,$24.99,auto
INSERT,SKU-NEW-002,Another New Product,A,Y,Tools,$49.99,new

```

Intent column behavior:

Intent	Behavior
INSERT	Insert new record. Fails if refName already exists.
UPDATE	Update existing record. Fails if refName does not exist.
UPSERT	Auto-detect: INSERT if new, UPDATE if exists. (Default behavior)

Intent	Behavior
SKIP	Skip this row entirely - no database operation.
MERGE	Not supported - returns error.
DELETE	Not supported - returns error.

3.6.4. API Calls

```
# Create the profile first
curl -X POST \
  -H "Authorization: Bearer $JWT" \
  -H "Content-Type: application/json" \
  "https://host/api/integration/import-profiles" \
  -d @product-import-profile.json

# Import using the profile (UPSERT behavior - auto-detect insert vs update)
curl -X POST \
  -H "Authorization: Bearer $JWT" \
  -F "file=@products.csv" \
  "https://host/api/products/csv?profileRefName=product-import-v1&skipHeaderRow=true"

# Import with explicit intent column (no profile needed for intent)
curl -X POST \
  -H "Authorization: Bearer $JWT" \
  -F "file=@products-with-intent.csv" \

"https://host/api/products/csv?requestedColumns=_action,refName,displayName,status&intentColumn=_action&skipHeaderRow=true"

# Import with intent column and profile for value mappings
curl -X POST \
  -H "Authorization: Bearer $JWT" \
  -F "file=@products-with-intent.csv" \
  "https://host/api/products/csv?profileRefName=product-import-v1&intentColumn=_action&skipHeaderRow=true"

# Import with default intent (all rows treated as INSERT, fail if exists)
curl -X POST \
  -H "Authorization: Bearer $JWT" \
  -F "file=@new-products.csv" \
  "https://host/api/products/csv?profileRefName=product-import-v1&defaultIntent=INSERT&skipHeaderRow=true"
```

3.6.5. Example PreValidationTransformer

```
package com.example.imports;

import com.e2eq.framework.util.csv.ImportRowContext;
```

```

import com.e2eq.framework.util.csv.PreValidationTransformer;
import com.e2eq.framework.util.csv.TransformationException;
import com.example.model.Product;
import jakarta.enterprise.context.ApplicationScoped;

import java.math.BigDecimal;

@ApplicationScoped
public class ProductTransformer implements PreValidationTransformer<Product> {

    @Override
    public Class<Product> getTargetType() {
        return Product.class;
    }

    @Override
    public int getPriority() {
        return 500; // Run before default transformers
    }

    @Override
    public Product transform(Product product, ImportRowContext context)
        throws TransformationException {

        // Compute derived fields
        if (product.getPrice() != null && product.getQuantity() != null) {
            product.setTotalValue(
                product.getPrice().multiply(BigDecimal.valueOf(product.getQuantity()))
            );
        }

        // Business rule: new products default to DRAFT status
        if (product.getStatus() == null) {
            product.setStatus(ProductStatus.DRAFT);
        }

        // Cross-field validation
        if (product.getDiscountPrice() != null && product.getPrice() != null) {
            if (product.getDiscountPrice().compareTo(product.getPrice()) > 0) {
                throw new TransformationException(
                    "discountPrice",
                    "Discount price cannot exceed regular price"
                );
            }
        }

        // Conditional transformation
        if ("CLEARANCE".equals(product.getCategory())) {
            product.setFeatured(false);
            product.setSearchable(false);
        }
    }
}

```

```
    return product;  
  }  
}
```

Chapter 4. Migration Path

4.1. Phase 1: Core Models (Non-Breaking)

1. Add `ImportProfile` entity and repository
2. Add `ColumnMapping`, `LookupConfig`, `GlobalTransformations`, `CaseTransform`, enums
3. Add `LookupService` interface
4. Create `ImportProfileResource` REST endpoint

Backward Compatibility: No changes to existing behavior.

4.2. Phase 2: Cell Processors (Non-Breaking)

1. Add `ValueMapProcessor`
2. Add `RegexReplaceProcessor`
3. Add `CaseTransformProcessor`
4. Add `LookupProcessor`
5. Add `EmptyToNullProcessor`

Backward Compatibility: New processors only used when profile specified.

4.3. Phase 3: CSVImportHelper Enhancement (Non-Breaking)

1. Add overloaded methods accepting `ImportProfile`
2. Add `buildProcessorsFromProfile()` method
3. Implement processor chain building from profile
4. Add `LookupService` implementation

Backward Compatibility: Existing methods unchanged; new methods are additions.

4.4. Phase 4: PreValidationTransformer SPI (Non-Breaking)

1. Add `PreValidationTransformer` interface
2. Add `ImportRowContext` class
3. Add `TransformationException`
4. Integrate transformer invocation into import flow

Backward Compatibility: Transformers are optional; empty by default.

4.5. Phase 5: Header Modifiers (Non-Breaking)

1. Add `HeaderModifier` enum and `ParsedHeader` class
2. Add `FieldCalculator` SPI interface
3. Add built-in calculators (Timestamp, UUID, RefName)
4. Add `InlineFieldCalculator` for profile-defined calculations
5. Add `enableHeaderModifiers` flag to `ImportProfile`
6. Add `enableHeaderModifiers` REST parameter
7. Integrate header parsing into import flow

Backward Compatibility: Feature is opt-in; disabled by default.

4.6. Phase 6: REST API Updates (Non-Breaking)

1. Add `profileRefName` query parameter to CSV endpoints
2. Add `intentColumn` and `defaultIntent` parameters
3. Add `enableHeaderModifiers` parameter
4. Update OpenAPI documentation
5. Add user guide documentation

Backward Compatibility: All parameters are optional; existing calls work unchanged.

Chapter 5. Security Considerations

5.1. Profile Access Control

- `ImportProfile` entities follow standard `DataDomain` scoping
- Profiles can only be used within their tenant/org context
- Cross-tenant profile access is blocked

5.2. Lookup Security

- Lookup queries respect the caller's security context
- `RuleContext` filters are applied to lookup queries
- Cross-tenant lookups are blocked by `DataDomain` filtering
- Lookups only access collections the user has `VIEW` permission on

5.3. Regex Safety

- Consider limiting regex complexity to prevent ReDoS attacks
- Implement timeout for regex operations on large values
- Log warnings for potentially dangerous patterns

Chapter 6. Performance Considerations

6.1. Lookup Caching

- Lookups are cached per-import session by default (`cacheLookups: true`)
- Cache is cleared between import sessions
- Consider LRU eviction for imports with many unique lookup values

6.2. Batch Lookup Optimization

- `LookupService.batchLookup()` allows batch resolution
- Collect all values for a column, resolve in single query
- Reduces database round-trips for large imports

6.3. Profile Caching

- Frequently used profiles could be cached in memory
- Consider cache invalidation on profile update

Chapter 7. Testing Strategy

7.1. Unit Tests

- Test each cell processor in isolation
- Test processor chain building with various configurations
- Test value mapping with case sensitivity options
- Test lookup with various failure behaviors
- Test PreValidationTransformer priority ordering
- Test header modifier parsing (*, ?, ~, #)
- Test ParsedHeader.parse() with various inputs
- Test FieldCalculator invocation for calculated fields
- Test required field validation
- Test key field intent detection

7.2. Integration Tests

- Test full import flow with profiles
- Test lookup resolution across collections
- Test security filtering on lookups
- Test error handling and reporting
- Test header modifiers with real CSV files
- Test backward compatibility (modifiers disabled)
- Test calculated fields with CDI-discovered calculators
- Test inline calculators from profile

7.3. Performance Tests

- Benchmark imports with 100K+ rows
- Measure lookup cache effectiveness
- Profile memory usage during large imports

Chapter 8. Appendix A: Common Value Mapping Patterns

8.1. Boolean Values

```
{
  "valueMappings": {
    "Y": "true", "N": "false",
    "YES": "true", "NO": "false",
    "TRUE": "true", "FALSE": "false",
    "1": "true", "0": "false",
    "T": "true", "F": "false",
    "ON": "true", "OFF": "false"
  },
  "valueMappingCaseSensitive": false
}
```

8.2. Status Codes

```
{
  "valueMappings": {
    "10": "DRAFT",
    "20": "PENDING",
    "30": "APPROVED",
    "40": "ACTIVE",
    "50": "SUSPENDED",
    "90": "CLOSED",
    "99": "CANCELLED"
  }
}
```

8.3. Country Codes

```
{
  "valueMappings": {
    "USA": "US",
    "United States": "US",
    "UK": "GB",
    "United Kingdom": "GB",
    "Great Britain": "GB"
  },
  "valueMappingCaseSensitive": false,
  "unmappedValueBehavior": "PASSTHROUGH"
}
```

}

Chapter 9. Appendix B: Regex Patterns

9.1. Remove Currency Symbols

```
{  
  "regexPattern": "[$€£¥,]",  
  "regexReplacement": ""  
}
```

9.2. Normalize Phone Numbers

```
{  
  "regexPattern": "[^0-9+]",  
  "regexReplacement": ""  
}
```

9.3. Extract Numeric ID

```
{  
  "regexPattern": "^([A-Z]+)-([0-9]+)$",  
  "regexReplacement": "$1"  
}
```

9.4. Normalize Whitespace

```
{  
  "regexPattern": "\\s+",  
  "regexReplacement": " "  
}
```

Chapter 10. Implementation Status

The following components have been implemented:

10.1. Completed Components

Component	Package	Description
CaseTransform	com.e2eq.framework.model.persistent.imports	Enum for string case transformation options
UnmappedValueBehavior	com.e2eq.framework.model.persistent.imports	Enum for behavior when CSV value not in mappings
LookupFailBehavior	com.e2eq.framework.model.persistent.imports	Enum for behavior when lookup fails
LookupConfig	com.e2eq.framework.model.persistent.imports	Configuration for cross-collection lookups
ColumnMapping	com.e2eq.framework.model.persistent.imports	Per-column transformation configuration
GlobalTransformations	com.e2eq.framework.model.persistent.imports	Global string transformation settings
InlineFieldCalculator	com.e2eq.framework.model.persistent.imports	Inline calculator definitions for profiles
HeaderModifier	com.e2eq.framework.model.persistent.imports	Enum for header modifier suffixes (*, ?, ~, #)
ParsedHeader	com.e2eq.framework.model.persistent.imports	Parsed CSV header with modifier
ImportIntent	com.e2eq.framework.model.persistent.imports	Import intent enum (INSERT, UPDATE, SKIP, UPSERT)
ImportProfile	com.e2eq.framework.model.persistent.imports	Persistent profile entity extending BaseModel
ImportSession (updated)	com.e2eq.framework.model.persistent.imports	Added <code>profileRefName</code> field
ImportProfileRepo	com.e2eq.framework.model.persistent.morphia	Repository for ImportProfile entities
FieldCalculator	com.e2eq.framework.imports.spi	SPI interface for field calculators
PreValidationTransformer	com.e2eq.framework.imports.spi	SPI interface for pre-validation transformers
ImportContext	com.e2eq.framework.imports.spi	Context object for SPI implementations
LookupService	com.e2eq.framework.imports.service	Interface for cross-collection lookups
LookupServiceImpl	com.e2eq.framework.imports.service	Default implementation with caching

Component	Package	Description
ImportProfileService	com.e2eq.framework.imports.service	Orchestration service for profile processing
TimestampFieldCalculator	com.e2eq.framework.imports.calculators	Built-in calculator for timestamp fields
UUIDFieldCalculator	com.e2eq.framework.imports.calculators	Built-in calculator for UUID fields
RefNameFieldCalculator	com.e2eq.framework.imports.calculators	Built-in calculator for refName generation
ValueMapProcessor	com.e2eq.framework.imports.processors	CellProcessor for value mapping
RegexReplaceProcessor	com.e2eq.framework.imports.processors	CellProcessor for regex replacement
CaseTransformProcessor	com.e2eq.framework.imports.processors	CellProcessor for case transformation
LookupProcessor	com.e2eq.framework.imports.processors	CellProcessor for cross-collection lookups
GlobalTransformProcessor	com.e2eq.framework.imports.processors	CellProcessor for global transformations
RowValueResolver	com.e2eq.framework.imports.spi	SPI interface for arbitrary per-row code execution
RowValueResolverProcessor	com.e2eq.framework.imports.processors	CellProcessor wrapper for RowValueResolver
CSVImportHelper (enhanced)	com.e2eq.framework.util	Added <code>analyzeCSVWithProfile()</code> method
BaseResource (enhanced)	com.e2eq.framework.rest.resources	Added <code>/csv/session/profile</code> endpoint

10.2. REST API Endpoints

Method	Path	Description
POST	<code>/[entity]/csv/session/profile</code>	Create import session with ImportProfile support. Accepts <code>profileRefName</code> query parameter.

10.3. Usage

To use the new profile-based import:

1. **Create an ImportProfile** via the standard BaseResource CRUD endpoints or directly in the database
2. **Call the profile endpoint:**

```
curl -X POST \  
  -H "Authorization: Bearer $TOKEN" \  
  -F "file=@data.csv" \  
  "https://host/api/products/csv/session/profile?profileRefName=my-profile"
```

3. **Review the preview** and commit using the existing session workflow:

```
# Commit the session  
curl -X POST \  
  -H "Authorization: Bearer $TOKEN" \  
  "https://host/api/products/csv/session/{sessionId}/commit"
```

Chapter 11. RowValueResolver: Per-Row Arbitrary Code Execution

The `RowValueResolver` SPI allows you to run arbitrary code for each row during CSV import. Unlike static lookups which match a single field against a collection, RowValueResolvers have access to **all columns in the row** and can invoke external services, apply complex business logic, or perform conditional transformations.

11.1. Use Cases

- **Geocoding addresses:** Look up latitude/longitude from a geocoding service based on address fields
- **Cross-field validation:** Validate combinations of fields that depend on each other
- **External service calls:** Call APIs to enrich, validate, or transform data
- **Complex business rules:** Apply conditional logic based on multiple fields
- **Composite key generation:** Generate identifiers from multiple source columns

11.2. Example: Address Geocoding

This example shows how to geocode addresses during import, setting latitude and longitude fields based on an address lookup service.

11.2.1. 1. Create the RowValueResolver Implementation

```
package com.example.imports;

import com.e2eq.framework.imports.spi.ImportContext;
import com.e2eq.framework.imports.spi.RowValueResolver;
import jakarta.enterprise.context.ApplicationScoped;
import jakarta.inject.Inject;
import jakarta.inject.Named;

import java.util.Map;

@ApplicationScoped
@Named("geocodeResolver")
public class GeocodeResolver implements RowValueResolver {

    @Inject
    GeocodingService geocodingService;

    @Override
    public String getName() {
        return "geocodeResolver";
    }
}
```

```

@Override
public ResolveResult resolve(String inputValue, Map<String, Object> rowData,
ImportContext context) {
    // Access multiple address columns from the row
    String street = getString(rowData, "street");
    String city = getString(rowData, "city");
    String state = getString(rowData, "state");
    String zip = getString(rowData, "zip");
    String country = getString(rowData, "country");

    // Build full address for geocoding
    String fullAddress = String.format("%s, %s, %s %s, %s",
        street, city, state, zip, country);

    try {
        // Call external geocoding service
        GeocodingResult result = geocodingService.geocode(fullAddress);

        if (result == null || !result.isValid()) {
            // Return null - field won't be set but row continues
            return ResolveResult.nullValue();
        }

        // Return a GeoPoint or coordinate object
        return ResolveResult.success(new GeoPoint(result.getLatitude(), result
.getLongitude()));

    } catch (GeocodingException e) {
        // Log warning but don't fail the row
        return ResolveResult.nullValue();
    } catch (Exception e) {
        // Critical error - skip this row
        return ResolveResult.error("Geocoding failed: " + e.getMessage());
    }
}

private String getString(Map<String, Object> rowData, String key) {
    Object value = rowData.get(key);
    return value != null ? value.toString().trim() : "";
}
}

```

11.2.2. 2. Configure the ImportProfile

```

{
  "refName": "location-import-profile",
  "displayName": "Location Import with Geocoding",
  "targetCollection": "com.example.model.Location",

```

```

"columnMappings": [
  {
    "sourceColumn": "name",
    "targetField": "displayName"
  },
  {
    "sourceColumn": "street",
    "targetField": "address.street"
  },
  {
    "sourceColumn": "city",
    "targetField": "address.city"
  },
  {
    "sourceColumn": "state",
    "targetField": "address.state"
  },
  {
    "sourceColumn": "zip",
    "targetField": "address.postalCode"
  },
  {
    "sourceColumn": "country",
    "targetField": "address.country",
    "defaultValue": "USA"
  },
  {
    "sourceColumn": "street",
    "targetField": "location",
    "rowValueResolverName": "geocodeResolver"
  }
]
}

```

Note: The `sourceColumn` for the `geocodeResolver` can be any column - the resolver has access to all row data via the `rowData` map. The column value is passed as `inputValue` but you can ignore it and use any combination of columns.

11.2.3. 3. Sample CSV Input

```

name,street,city,state,zip,country
"Acme Corp HQ","123 Main St","San Francisco","CA","94102","USA"
"East Coast Office","456 Park Ave","New York","NY","10022","USA"
"European Branch","10 Oxford St","London","","W1D 1BS","UK"

```

11.2.4. 4. Result

After import, each Location entity will have:

- All address fields populated from the CSV
- A **location** field containing a GeoPoint with latitude/longitude

11.3. RowValueResolver API

```
public interface RowValueResolver {

    /**
     * Unique name for this resolver (used in ColumnMapping.rowValueResolverName).
     */
    String getName();

    /**
     * Resolve a value based on input and row data.
     *
     * @param inputValue the value from the CSV column (after prior transformations)
     * @param rowData all column values from the current row (column name -> value)
     * @param context import context with profile, realm, row number, session ID
     * @return the resolution result
     */
    ResolveResult resolve(String inputValue, Map<String, Object> rowData,
        ImportContext context);

    /**
     * Check if this resolver applies to the given target class.
     * Default returns true (applies to all types).
     */
    default boolean appliesTo(Class<?> targetClass) {
        return true;
    }

    /**
     * Priority order - lower values run first. Default is 100.
     */
    default int getOrder() {
        return 100;
    }
}
```

11.4. ResolveResult Options

Method	Description
<code>ResolveResult.success(value)</code>	Return a resolved value to set on the target field
<code>ResolveResult.passthrough(originalValue)</code>	Keep the original value unchanged

Method	Description
<code>ResolveResult.nullValue()</code>	Set the field to null (but continue processing)
<code>ResolveResult.skip()</code>	Skip this row entirely (not an error)
<code>ResolveResult.skip(reason)</code>	Skip with a reason message
<code>ResolveResult.error(message)</code>	Mark row as error with the given message

11.5. Processing Order

RowValueResolvers are applied after field calculators but before pre-validation transformers:

1. CSV parsing with CellProcessors (value mapping, regex, case transform, static lookups)
2. Field calculators (timestamps, UUIDs, templates)
3. **RowValueResolvers** (arbitrary code with full row access)
4. Pre-validation transformers
5. Bean validation